



Secure Exit

SIMATS Online Programming Lab

JAVA PROGRAMMING

Delisha A K
192511307RC

QUESTIONS

Q1
Student Management System
[BL3 – Apply]
10 marks

Q2
2. Library Book Management
[BL3 – Apply]
10 marks

Q3
Employee Salary Calculation
[BL3 – Apply]
10 marks

Q4
Shape Area Calculator using
Polymorphism [BL3 – Apply]
10 marks

Q5
Vehicle Registration System
[BL3 – Apply]
10 marks

Q6
Bank Account Hierarchy [BL4 –
Analyze]
10 marks

Q7
Hospital Patient Record System
[BL4 – Analyze]
10 marks

Q8
E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

Q1 Student Management System [BL3 – Apply] 10 marks

Write pseudocode to design a Student class with attributes (name, rollNo, marks) and methods to:

- Accept student details
- Calculate average marks
- Display result (Pass/Fail)

Apply encapsulation and data hiding by restricting direct access to attributes.
OOP Concepts: Encapsulation | Data Hiding | Classes & Objects

Your Answer

```
START
CLASS Student
PRIVATE name, rollno, marks
METHOD setDetails()
READ name, rollno, marks
METHOD calculateaverage()
average=marks
METHOD displayresult()
IF average>=50
PRINT "pass"
else
PRINT "fail"
END CLASS

CREATE student s
CALL setdetails()
CALL calculateaverage()
CALL displayresult()

STOP
```

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10 marks

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Q2 2. Library Book Management [BL3 – Apply] 10 marks

Design a Book class with private attributes (title, author, isbn, isAvailable). Implement methods to:

- Issue a book (change availability status)
- Return a book
- Display book details using getter methods

Demonstrate object creation for at least 3 books and simulate issue/return operations.
OOP Concepts: Encapsulation | Getters & Setters | Objects

Your Answer

```
START
CLASS book
PRIVATE title,author,isbn,
METHOD issuebook()
isavailable=false
METHOD returnbook()
isavailable=true
METHOD getdetails()
PRINT title ,author,isbm
END CLASS

CREATE book b1,b2,b3
CALL b1.issuebook()
CALL b2.returnbook()
CALL b1.getdetails()

STOP
```

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Q9

Q3 Employee Salary Calculation [BL3 – Apply]

10 marks

Write pseudocode for an Employee class with encapsulated attributes (empId, name, basicPay, designation). Implement methods to:

- Calculate gross salary (basic + HRA + DA)
- Calculate net salary after deductions (PF, tax)
- Display pay slip

Apply data hiding so salary components are not directly accessible.
OOP Concepts: Encapsulation | Data Hiding | Methods

Your Answer

```
START
CLASS employee
PRIVATE empId,name,basicpay,
METHOD calcsalary()
gross=basicpay+hra+da
net=gross-pf-tax
METHOD displaypayslip()
PRINT empId,name,gross,net
END CLASS
CREATE employee e
CALL calcsalary()
CALL displaypayslip()
STOP
```

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E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9

Q4 Shape Area Calculator using Polymorphism [BL3 – Apply]

10 marks

Create a Shape base class with an overridable calculateArea() method. Apply this to:

- Circle — area using radius
- Rectangle — area using length and breadth
- Triangle — area using base and height

Use method overriding to demonstrate runtime polymorphism. Call calculateArea() for each object.
OOP Concepts: Inheritance | Polymorphism | Method Overriding

Your Answer

```
START
METHOD calculatearea()
CLASS circle EXTENDS shape
OVERRIDE calculatearea()
CLASS rectangle EXTENDS shape
OVERRIDE calculatearea()
CLASS triangle EXTENDS shape
OVERRIDEncalculatearea()
CREATE circle c
CREATE rectangle r
CREATE triangle t
CAL c.calculatearea()
CAL r.calculatearea()
CAL t.calculatearea()
STOP
```

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Q9

Q5 Vehicle Registration System [BL3 – Apply]

10 marks

Design a Vehicle class with attributes (regNo, owner, fuelType). Extend it into:

- TwoWheeler — with engine capacity
- FourWheeler — with seating capacity and AC status

Apply inheritance to reuse common attributes and override a displayDetails() method in each subclass.
OOP Concepts: Inheritance | Method Overriding | Subclass

Your Answer

```
START
CLASS vehicle
regno,owner,fueltype
METHOD displaydetails()
CLASS two Wheeler EXTENDS vehicle
OVERRIDE displaydetails()
CLASS four Wheeler EXTENDS vehicle
OVERRIDE displaydetails()
CREATE two Wheeler t
CREATE four Wheeler f
CALL t.displaydetails()
CALL f.displaydetails()
STOP
```

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10 marks

Q9

Q6 Bank Account Hierarchy [BL4 – Analyze]

10 marks

Develop pseudocode for a BankAccount superclass and two subclasses — SavingsAccount (with interest calculation) and CurrentAccount (with overdraft facility). Analyze and implement:

- Inheritance of common attributes (accNo, balance, holder)
- Method overriding for withdrawal behavior
- How overdraft protection differs from savings limit

Compare the behavior of withdraw() across both subclasses and justify the design choices.
OOP Concepts: Inheritance | Method Overriding | Polymorphism

Your Answer

```
START
CLASS bankaccount
accno,balance,holder
METHOD withdraw()
CLASS savingsaccount EXTENDS bankaccount
OVERRIDE withdraw()
METHOD interest()
CLASS currentaccount EXTENDS bankaccount
OVERRIDE withdraw()
METHOD overdraft()

CREATE savingsaccount s
CREATE currentaccount c
CALL s.withdraw()
CALL c.withdraw()
STOP
```

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Vehicle Registration System [BL3 – Apply]

10 marks

Q6

Bank Account Hierarchy [BL4 – Analyze]

10 marks

Q7

Hospital Patient Record System [BL4 – Analyze]

10 marks

Q8

E-Commerce Product Catalog [BL4 – Analyze]

10 marks

Q9

Q7 Hospital Patient Record System [BL4 – Analyze]

10 marks

Design a Patient base class and analyze how it extends into:

- InPatient — with ward number, admission date, and daily charges
- OutPatient — with appointment date and consultation fee

Analyze encapsulation requirements for sensitive data (diagnosis, prescription). Override a generateBill() method for each type and compare the billing logic differences. OOP Concepts: Encapsulation | Inheritance | Polymorphism

Your Answer

```
START
CLASS patient
PRIVATE diagnosis,prescription
METHOD generatebill()
CLASS inpatient EXTENDS patient
OVERRIDE generatebill()
CLASS outpatient EXTENDS patient
OVERRIDE generatebill()
CREATE inpatient i
CREATE outpatient o
CALL i.generatebill()
CALL o.generatebill()
STOP
```

Save Answer

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Student Management System [BL3 – Apply]

10 marks

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10 marks

Q3

Employee Salary Calculation [BL3 – Apply]

10 marks

Q4

Shape Area Calculator using Polymorphism [BL3 – Apply]

10 marks

Q5

Vehicle Registration System [BL3 – Apply]

10 marks

Q6

Bank Account Hierarchy [BL4 – Analyze]

10 marks

Q7

Hospital Patient Record System [BL4 – Analyze]

10 marks

Q8

E-Commerce Product Catalog [BL4 – Analyze]

10 marks

Q9

Q8 E-Commerce Product Catalog [BL4 – Analyze]

10 marks

Analyze and design a Product class hierarchy for an e-commerce system:

- Electronics — with warranty, brand, voltage
- Clothing — with size, fabric, gender
- Grocery — with expiryDate and weight

Override an applyDiscount() method for each category with different discount rules. Analyze why polymorphism makes the cart checkout process flexible and extensible. OOP Concepts: Polymorphism | Inheritance | Method Overriding

Your Answer

```
START
CLASS applydiscount()
CLASS electronics EXTENDS product
OVERRIDE applydiscount()
CLASS clothing EXTENDS product
OVERRIDE applydiscount()
CLASS grocery EXTENDS product
OVERRIDE applydiscount()
CREATE electronics e
CREATE clothing c
CREATE grocery g
CALL e.applydiscount()
CALL c.applydiscount()
CALL g.applydiscount()
STOP
```

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Q8
E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q9
University Staff Hierarchy [BL4 –
Analyze]
10 marks

Q10
Smart Home Device Controller
[BL4 – Analyze]
10 marks

10/10 answered

OOP Concepts: Multilevel Inheritance | Access Modifiers | Data Hiding

Your Answer

START
CLASS staff
PROTECTED calculateallowance()
CLASS teachingstaff EXTENDS staff
OVERRIDE calculateallowance()
CLASS professor EXTENDS teachingstaff
OVERRIDE calculateallowance()
CLASS lecturer EXTENDS teachingstaff
OVERRIDE calculateallowance()
CLASS nonteachingstaff EXTENDS staff
OVERRIDE calculateallowance()
CREATE professor p
CREATE lecturer l
CREATE nonteachingstaff n

CALL p.calculateallowance()
CALL l.calculateallowance()
CALL n.calculateallowance()
STOP

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Hospital Patient Record System
[BL4 – Analyze]
10 marks

Q8
E-Commerce Product Catalog
[BL4 – Analyze]
10 marks

Q10 Smart Home Device Controller [BL4 – Analyze] 10 marks

Analyze and design a SmartDevice superclass with subclasses: SmartLight, SmartThermostat, and SmartLock. Override an executeCommand(cmd) method in each, where the same command (e.g., 'on', 'off', 'status') produces device-specific behavior. Analyze:

- How polymorphism allows a single controller loop to manage all devices
- Security implications of data hiding in SmartLock (PIN must never be exposed) Design the class hierarchy and explain the OOP principles applied at each level.

OOP Concepts: Polymorphism | Encapsulation | Inheritance

Your Answer

START
CLASS smartdevice
METHOD executecommand()
CLASS smartlight EXTENDS smartdevice
OVERRIDE executecommand()
CLASS smarthermostat EXTENDS smartdevice
OVERRIDE executecommand()
CLASS smartlock EXTENDS smartdevice
PRIVATE PIN
OVERRIDE executecommand()
CREATE smartlight l
CREATE smarthermostat t
CREATE smartlock s
CALL l.executecommand()
CALL t.executecommand()
CALL s.executecommand()
STOP

Save Answer